UNIVERSITÀ DEGLI STUDI DI MILANO

DATA INTEGRATION WITH MOFA

Course: Scientific Visualization

Corti Filippo
Dal Santo Giorgio
Donato Carlotta

Code and Visualizations available at
https://github.com/Filippo-Corti/ScientificVisualization/tree/main/02-data-integration

HPO-GO Dataset Overview

4 Binary Matrices:

- **GO (Gene Ontology):** describes each Gene based on:
  - **CC** (Cellular Components): "Where" the Gene is.
  - **BP** (Biological Processes): "What" the Gene does (at a high level).
  - **MF** (Molecular Functions): "What" the Gene does (at a low level).

- **HPO (Human Phenotype Ontology):** describes Phenotypic Abnormalities:
  - What Symptoms does a Mutation in this Gene have?

For each matrix:

- **rows** = Genes (5183, the same for each view)
- **columns** = ontology terms (24794 across the 4 views)
- **entries** = 0/1 (not annotated/annotated)

Our Goals

Our goals are:

- Experiment with **Data Integration**, understanding:
  - In which way a **Model-Based Approach** is more robust than an **Early Integration**.
  - How to Evaluate the result of a Model-Based Approach (**MOFA**).

- Use the Integrated Data to:
  - Visualize the Dataset using **Data Integration-Specific Visualizations**.
  - Obtain a better **Understanding** of the Dataset through Visualization.

MOFA – Multi-Omics Factor Analysis

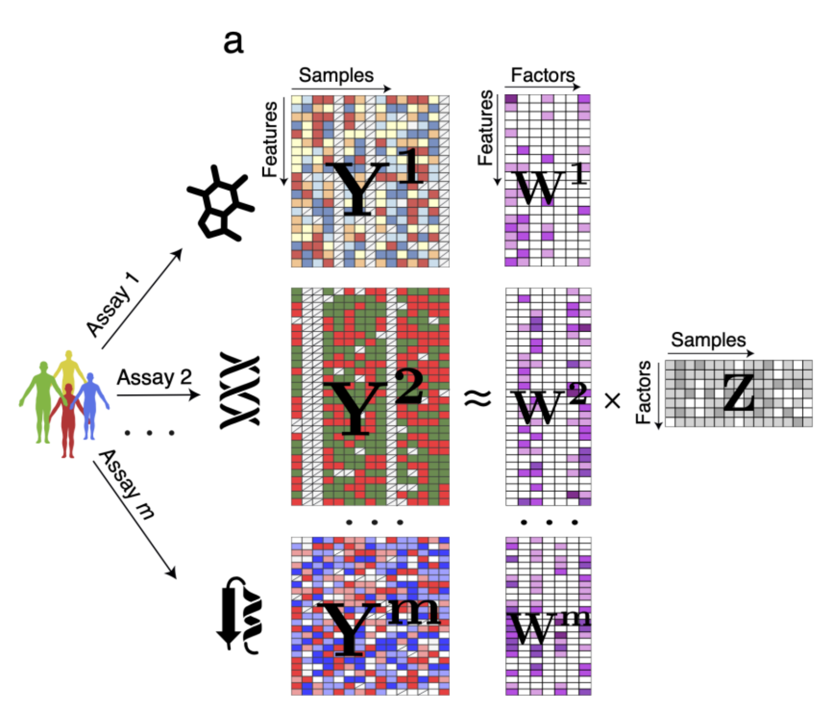
MOFA constitutes a **Model-Based Approach** to Data Integration:

- **Inputs**:
  - Multiple **-Omic Data** Types on the same Set of Samples (or on Overlapping Subsets).
- **Outputs**:
  - An **Interpretable Low-Dimensional Representation** of the Inputs.

It does so by identifying **Sources of Variation**, specific to one View or shared cross multiple ones.

In our Dataset:

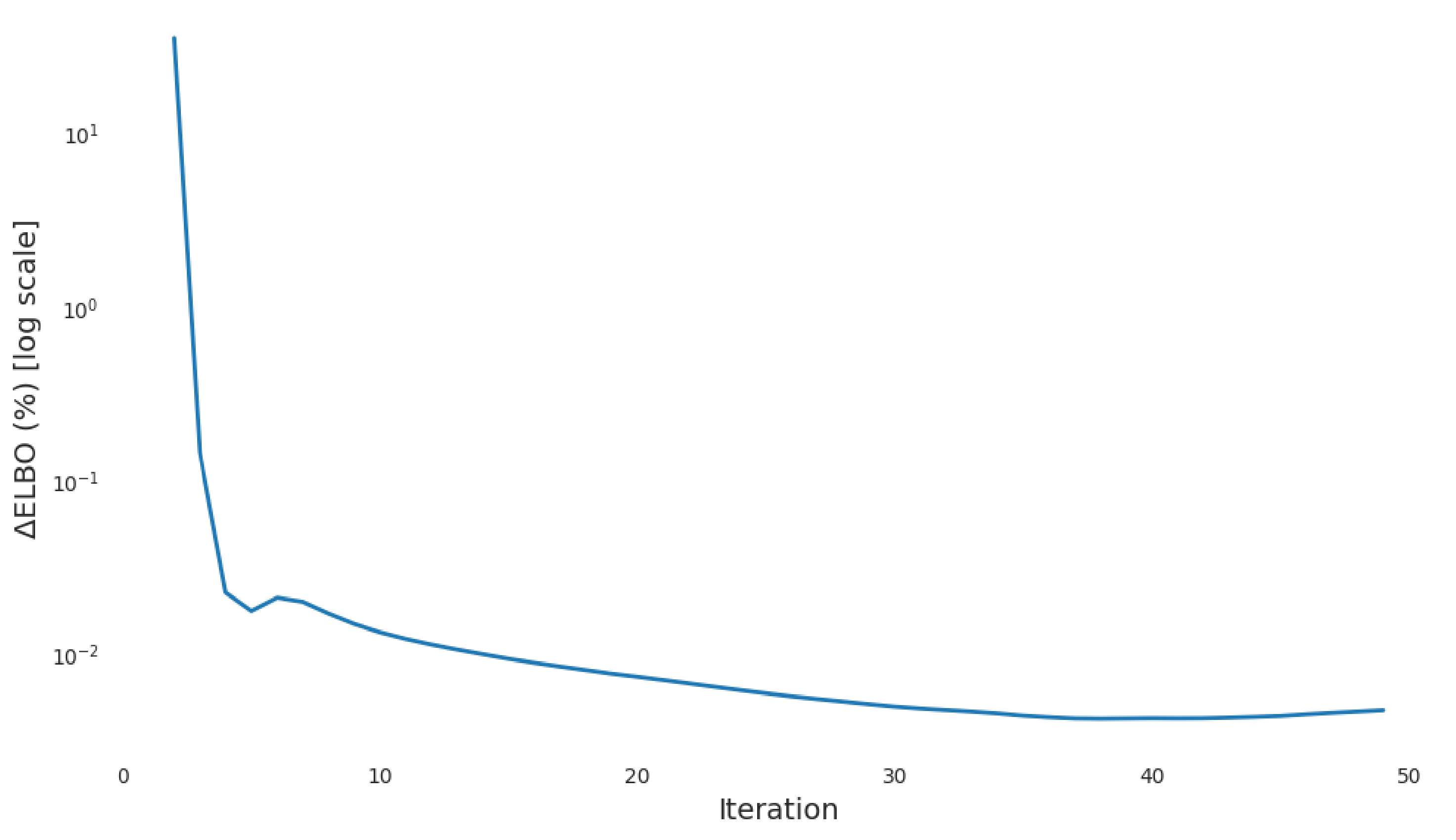
- It captures a **Global Structure** across GO and HPO Annotations.
- It produces a single **Latent Factor Matrix** of Size: N_Factors x N_Samples



MOFA – Training

We ran **MOFA** with the following parameters:

- 4 Views (5183 Samples):
  - **GO-BP: 9873** Features
  - **GO-CC: 1478** Features
  - **GO-MF: 3258** Features
  - **HPO: 10185** Features.
- **Bernoulli** Likelihoods
- **50** Iterations
- **15** Factors

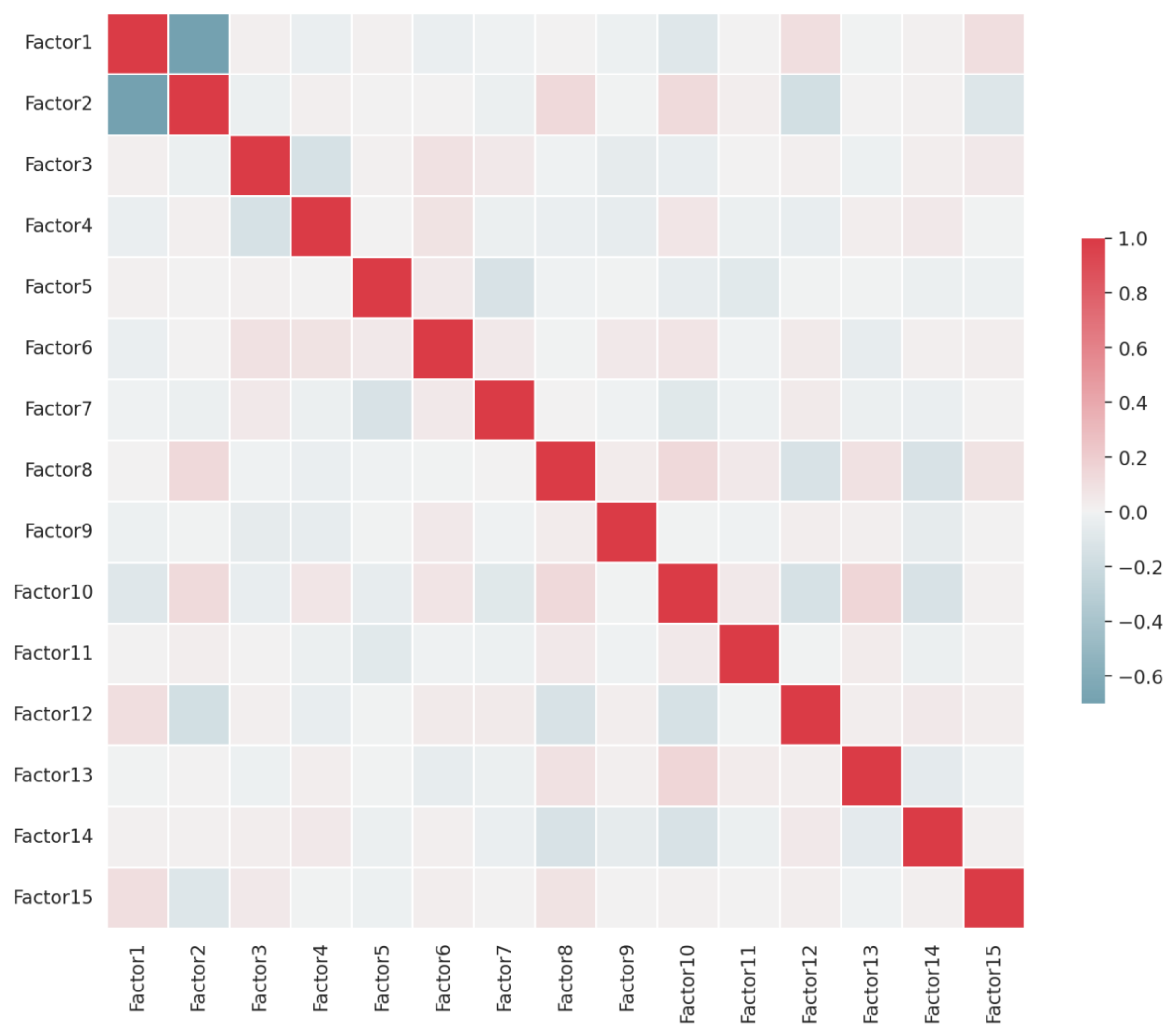


MOFA – Number of Factors

How do we choose **15 Factors**?

- The more Factors there are, the less **Interpretable** the Model becomes.
- Computing **Correlation** between the Factors can suggest that certain Factors are **copies/mirrors** of other Factors:
  - In our case, **Factor 1** and **Factor 2** appear highly **inversely correlated.**

Maybe we only need 1?



MOFA – Clustering

We Clustered the Samples (Genes):

- Using the Model from **MOFA**
  - **KMeans** for k = 2..100

- On each of the Views (GO-BP, GO-CC, GO-MF, HPO):
  - **MiniBatchKMeans** for k = 2..100
    - This is not what we would expect, considering Binary Data.
    - However, KModes would run way too slow.
    - And KMeans is actually considered to be approximating KModes fairly well for high-dimensionality Binary Data.

Result: 495 ways of Clustering the Samples

MOFA – Clustering & NMI

Two **Clustering Assignments** can be compared using **Normalized Mutual Information:**

$$\mathrm{MI}(U,V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} P(i,j) \log\left(\frac{P(i,j)}{P(i)P'(j)} \right)$$

$$\mathrm{NMI}(U,V) = \frac{\mathrm{MI}(U,V)}{\mathrm{mean}(H(U), H(V))}$$

Intuitively, it measures how much one Cluster tells you about the other.
In our case, we can compute a **495x495 NMI Matrix.**

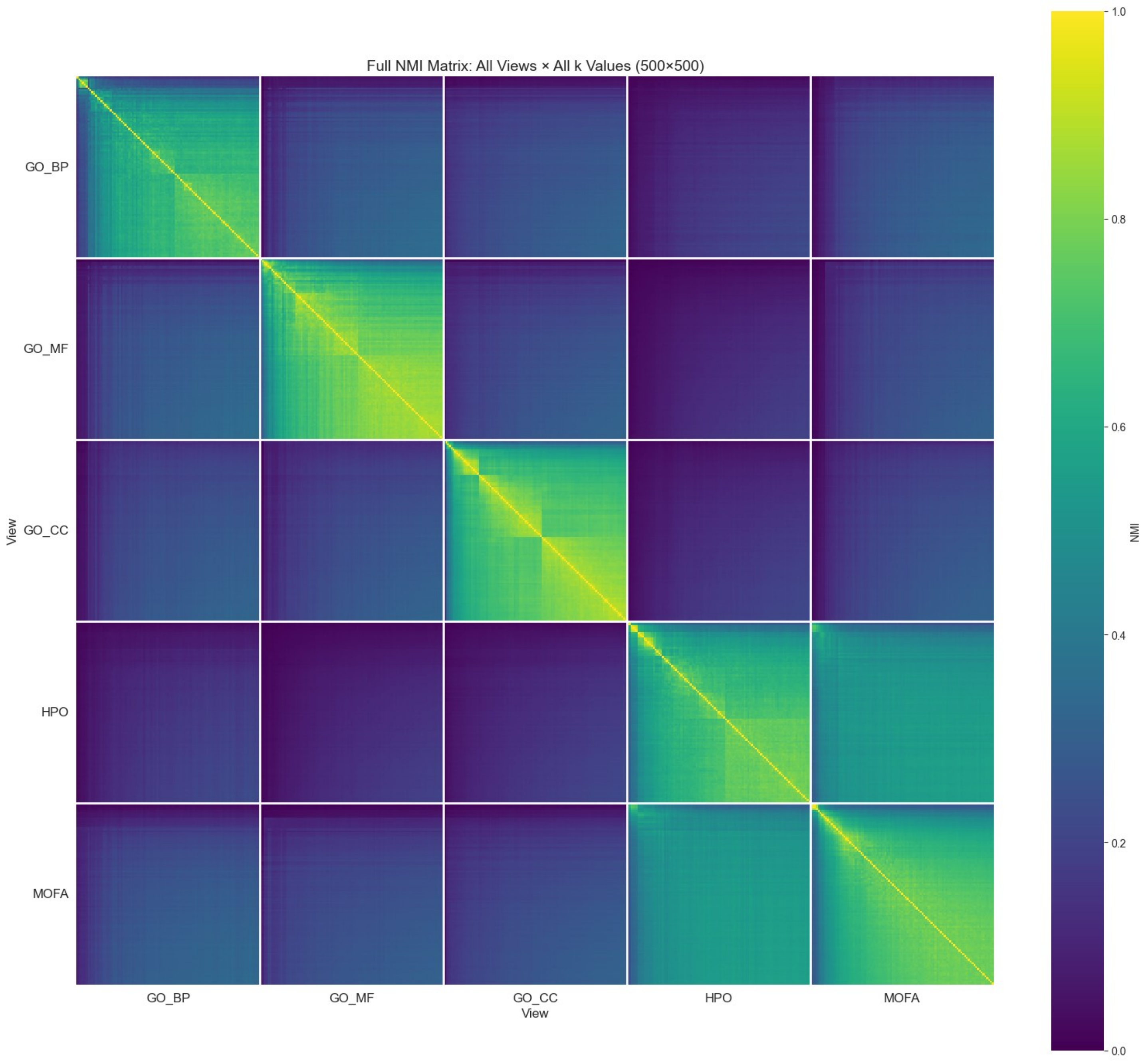
MOFA – NMI

Some observations:

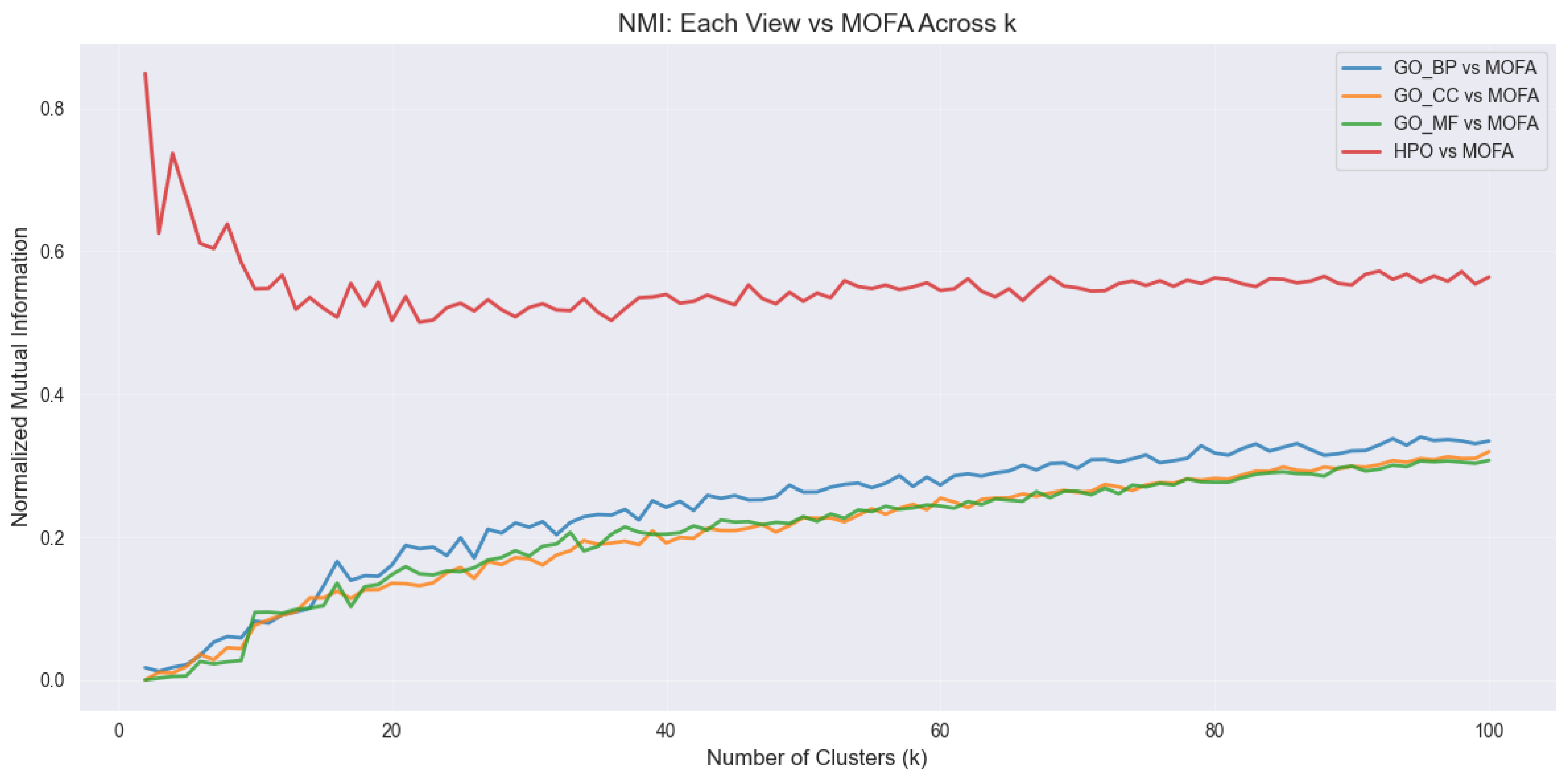
- Each **View-Pair Comparison** appears like a Color **Gradient.**

- Each View tends to **agree** more with **itself.**

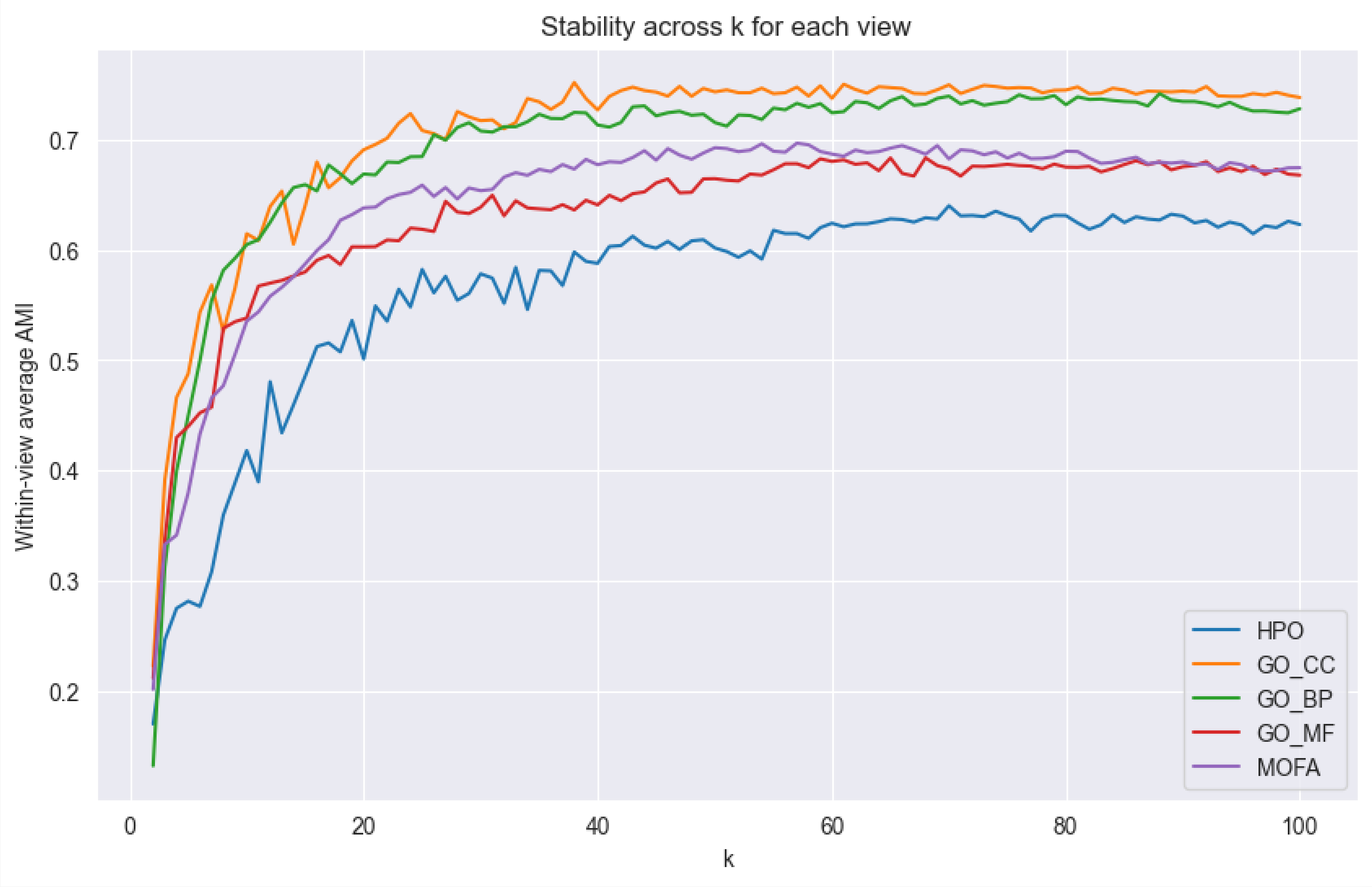
- **HPO** and **MOFA** are unusually **similar.**



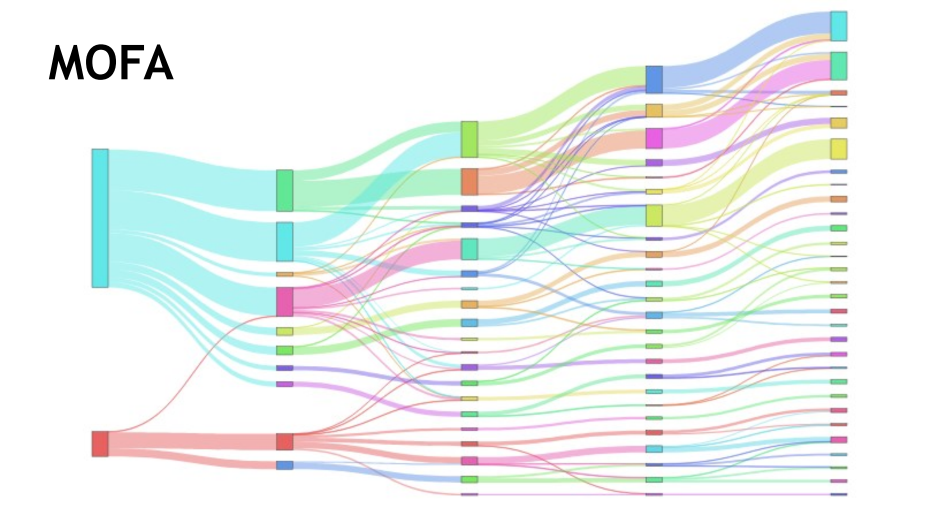
MOFA – NMI over different k's

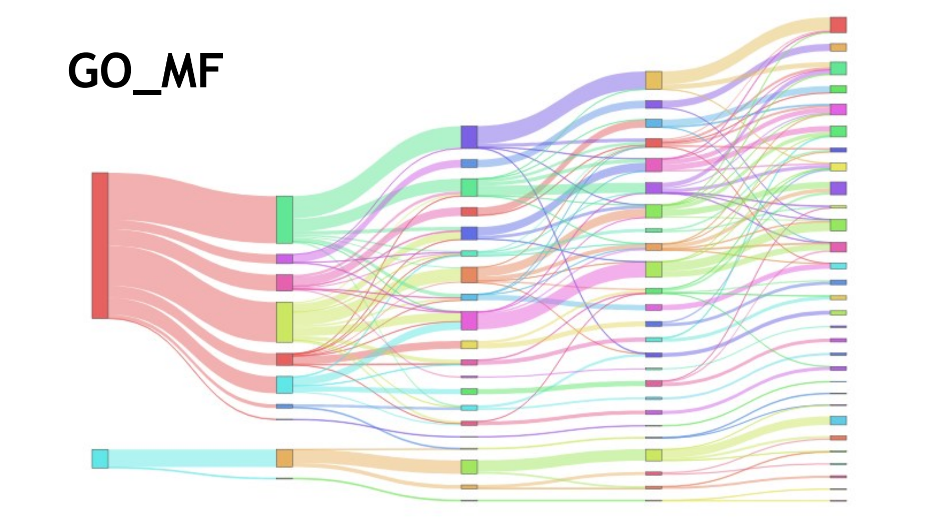


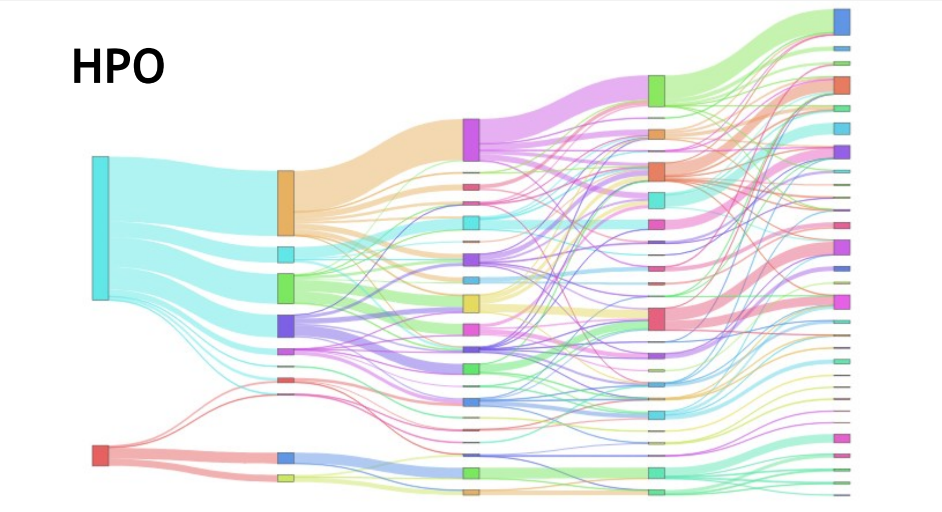
MOFA – NMI Stability over different k's



Sankey Plots – k=2, 10, 20, 25, 30







Notes

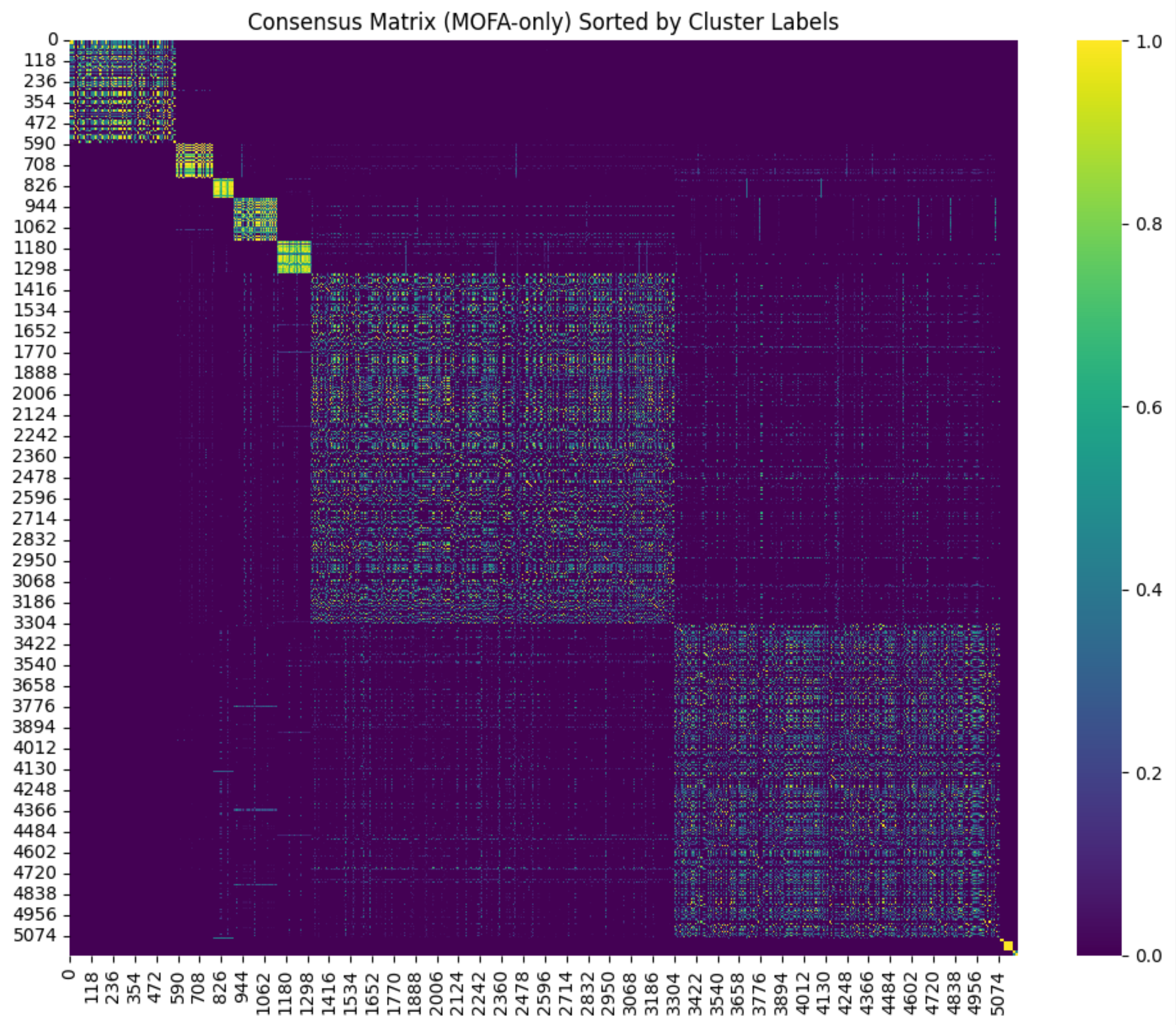
1. We computed the **consensus matrix** (5138×5138).
   - It represents, for each pair of samples, how often they end up in the same cluster.
2. We computed the **CDF** (Cumulative Distribution Function) of this matrix:
   - for a given consensus value (e.g., 0.8), it tells how many sample pairs have a value ≤ that threshold.
3. We computed the **Area Under the CDF** (AUC), which itself behaves like a cumulative stability curve.
4. For each value of k in the interpretable range (2–20), the AUC provides an overall measure of **clustering stability**.
5. We computed the ΔCDF and based on the stability jump we selected **k = 10**.
   - Each cluster is composed of a group of samples.
6. Using **G.Profiler**, we extracted the columns/features that typically characterize the samples in each cluster.
   - For each term (BP, CC, MF), we also obtain a **descriptive explanation.**
7. This gives us a hierarchy of grouped columns, aligned with the structure of the k = 10 clusters.
   - Each enriched term comes with its textual description.
8. At this point, **we can plot the graph**.

Consensus Clustering

- Consensus clustering asks: across all clustering results I have, **which genes repeatedly end up together**?

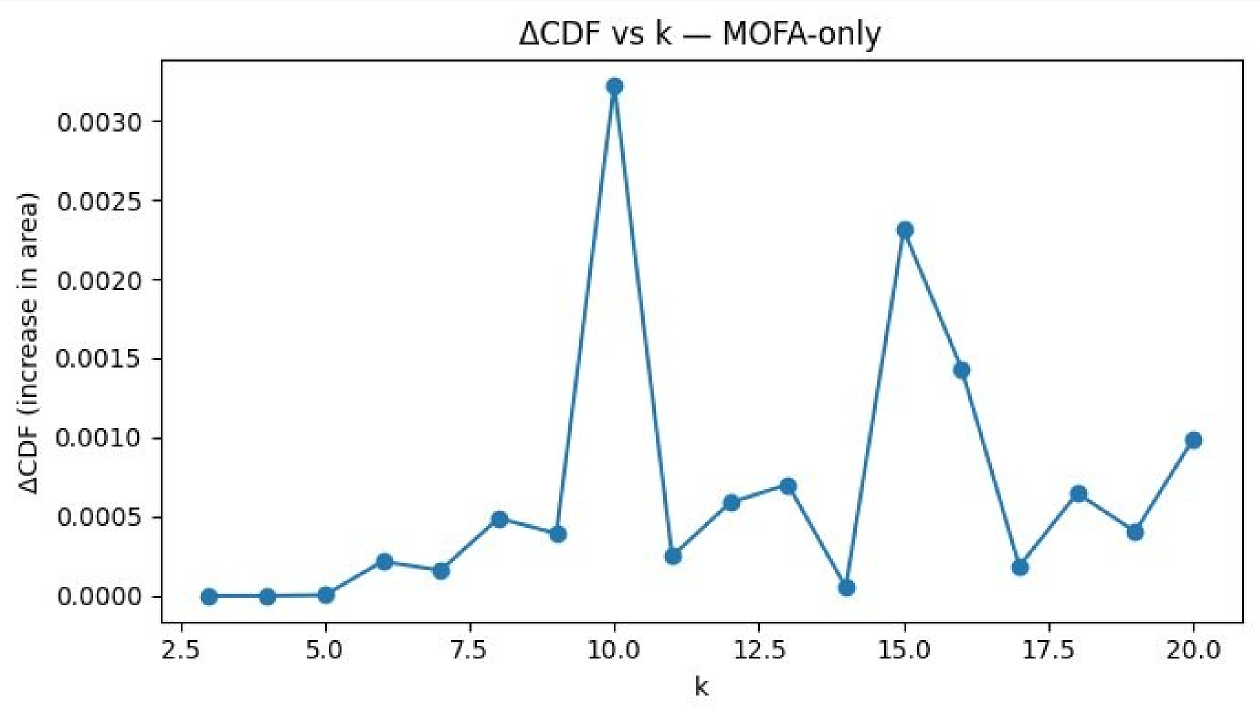
- It combines results from **multiple clustering runs** and identifies **stable and reproducible clusters.**

- Consensus clustering builds a **consensus matrix**, which is a sample-by-sample matrix. Each entry (i, j) means:
  - **1.0** → samples always end up in the same cluster across repeated clustering cycle.
  - **0.0** → samples never cluster together.



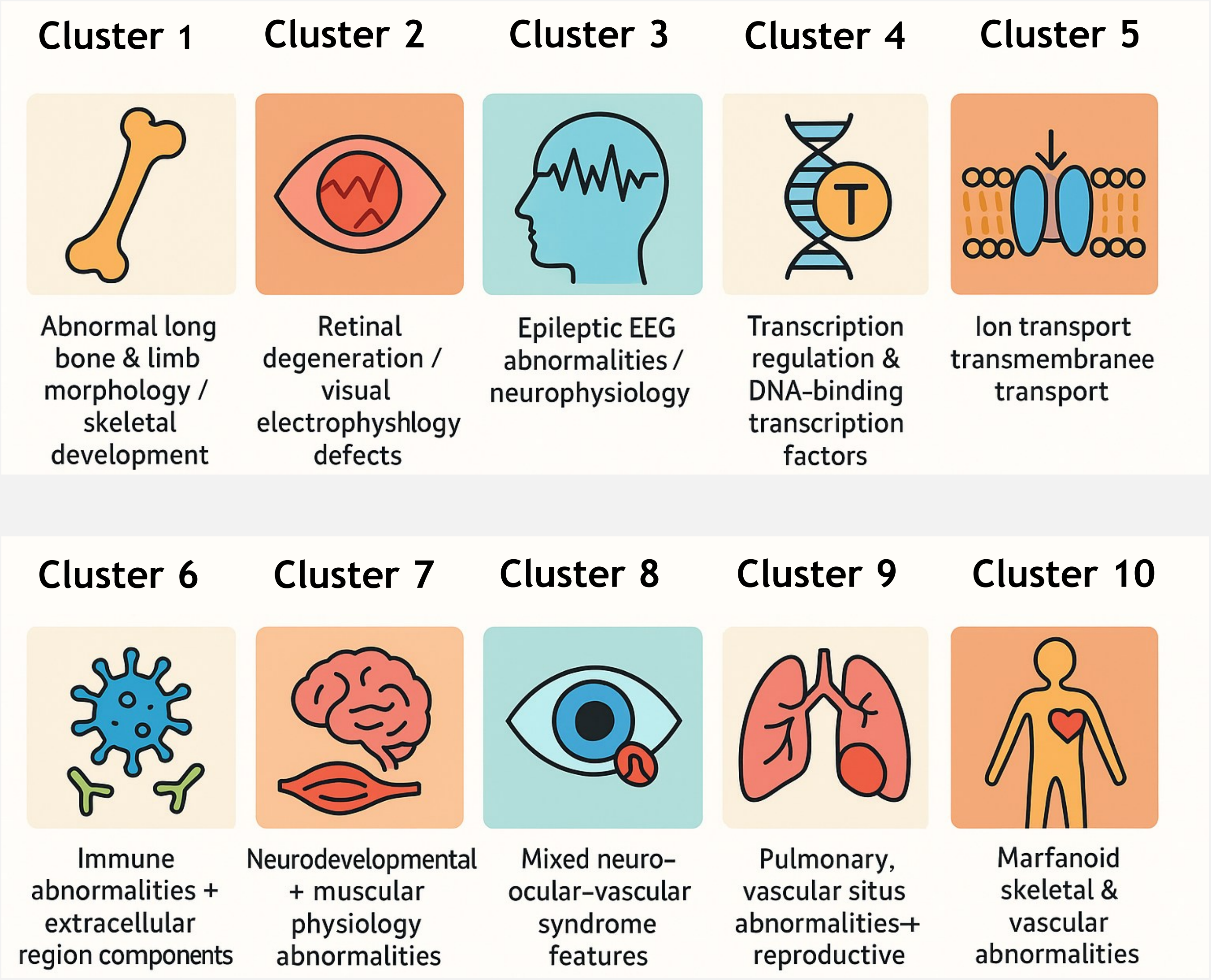
Consensus Clustering

- Build the **CDF curve**: **CDF(x)** = proportion of sample pairs with consensus ≤ x
- Calculate the **Area Under the CDF (AUC)**
- Compute **ΔCDF(k)**:
  - **Large ΔCDF** → adding clusters improves stability.
  - **Small ΔCDF** → additional clusters not meaningful.
- We selected **k = 10**

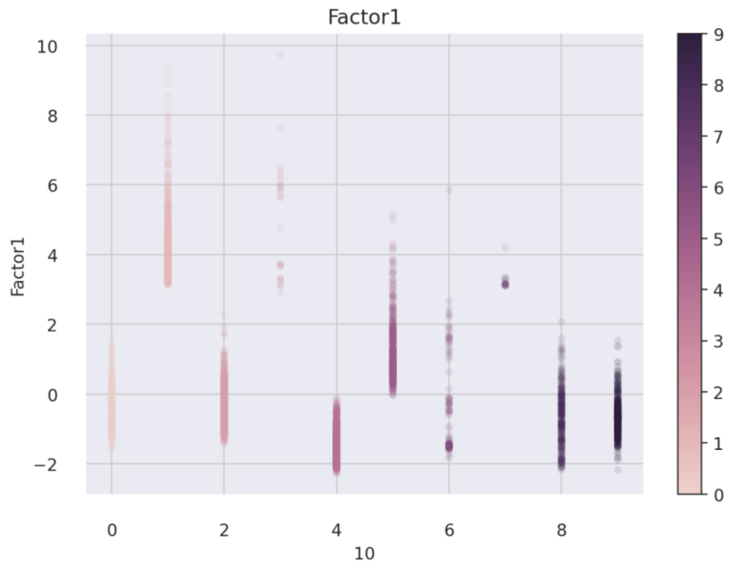


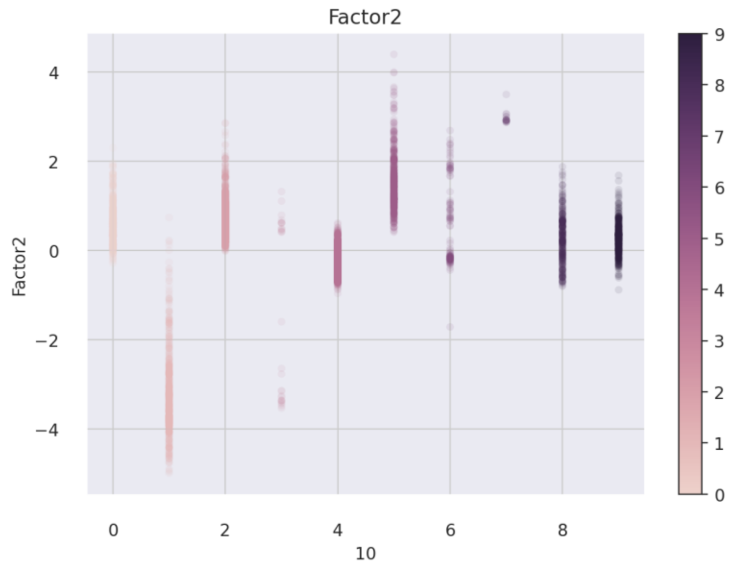
Functional Annotation of Consensus Clusters

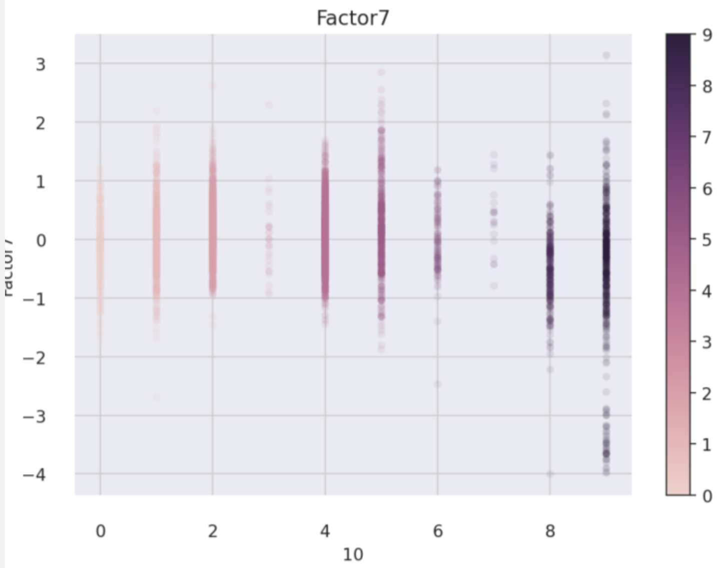
- After obtaining **consensus gene clusters**, we assign **biological meaning** using functional enrichment.
- ***g:Profiler*** tests whether each cluster's genes are enriched for known biological categories.
- For each cluster:
  - Input: genes in the cluster
  - Background: all measured genes
  - Output: significant functional terms
- From the enrichment table:
  - Keep **GO:BP** entries
  - Select the **top N most significant terms**

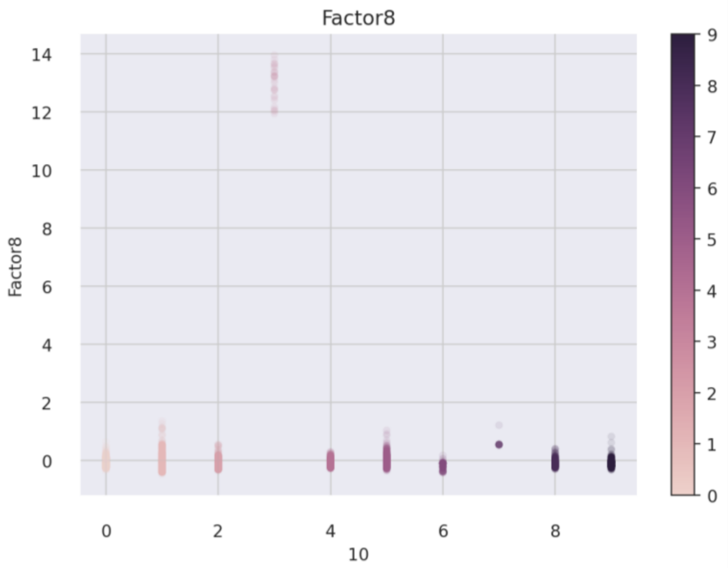


MOFA – Factors by Cluster









GO Biological Process Graph

Start from **top enriched GO:BP terms** per cluster:

- **Node size:** reflects how strongly enriched that GO term is in the cluster.
- **Edges:** represent GO hierarchy.
- **Node color:** indicates cluster membership.